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DataPollex Ltd. — Industrial Training Recruitment Process

Recruitment Task

Simplified Learning Management System (LMS)

Duration: 2 Days (48 Hours)
Project Type: Full-Stack Web Application
Technology: MERN or PERN Stack

1. Project Overview

Develop a simplified Learning Management System (LMS) with two panels:

- **Admin Dashboard:** Manage courses and lectures.
- **Student Panel:** Browse courses and view course content.

The objective is to evaluate candidates' practical knowledge of frontend development, backend APIs, database operations, and project structure.

2. Technical Requirements

Candidates may use either of the following stacks:

- **Frontend:** React or Next.js with TypeScript preferred.
- **Backend:** Node.js and Express.js.
- **Database:** MongoDB or PostgreSQL.
- **Styling:** Tailwind CSS or any suitable CSS framework.

The backend must follow a basic MVC structure, with separate routes, controllers, and models.

3. Admin Dashboard

A. Course Management

- Create a course with a title, description, price, and thumbnail URL.
- Display all courses in a card or table layout.
- Edit and delete existing courses.

B. Lecture Management

- Add lectures to a selected course.
- Each lecture should include a title, description, and video URL.
- Display lectures associated with their respective courses.

- Edit and delete lectures.

C. Basic Access Control

- Provide a simple, working admin login.
- Restrict course and lecture management to authenticated admins.

4. Student Panel

A. Course Listing

- Display all courses created by the admin.
- Show the course title, thumbnail, description, and price.
- Allow students to open a course details page.

B. Course Details and Lectures

- Display the selected course's information.
- List all lectures associated with that course.
- Allow students to open a lecture and watch its video using an embedded video URL.

Note: Student registration, payment, enrollment, PDF uploads, and advanced progress tracking are not required.

5. UI/UX Requirements

- Responsive design for desktop and mobile devices.
- Clear navigation between the course listing, course details, and admin dashboard.
- Basic loading, empty, and error states where appropriate.
- Consistent layout and readable interface.

6. Mandatory Technical Requirements

- Working frontend-to-backend API integration.
- Database persistence for courses and lectures.
- Proper CRUD operations.
- Basic input validation and error handling.
- Admin authentication and protected management routes.
- A modular backend structure following MVC principles.
- All student-facing course and lecture data must come from the database, not hardcoded frontend data.

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7. Submission Requirements

Candidates must submit:

1. **Source Code:** GitHub or GitLab repository.
2. **Live Demo:** A deployed application, if feasible within the deadline.
3. **Test Credentials:** Admin/User login credentials for evaluation.
4. **Database Setup:** Seed data or instructions to create initial data.

If deployment becomes a blocker, candidates must still submit a working local project with clear setup instructions.

8. Important Instructions

- Complete the task within **2 days of receiving it**.
- Focus on working core features rather than unnecessary complexity.
- Use of documentation and online references is permitted.
- Candidates should be prepared to explain their code and technical decisions during evaluation.
- Any incomplete feature should be clearly documented.

Submission Deadline: Within 2 days of receiving the task.

Submission Method: Submit the repository and demo details through the submission link.

https://forms.gle/N1js1UmWe69UDVrB7